NETWORK SCANNING USING NMAP

A Practical Documentation Using Kali Linux

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Course	MCA — Computer Applications / Cyber Security
Tool Used	Kali Linux and Nmap 7.99
Project Type	Network Discovery and Security Assessment
Evidence	Student-captured terminal screenshots
Project Date	September 2026

Lab note: Scanning should be limited to systems and networks that are owned by the student or explicitly authorized for security testing.

TABLE OF CONTENTS

1. Introduction 3
2. Project Objective 4
3. Scope 5
4. Tools Required 6
5. Install Nmap and Verify Setup 7
6. Host Discovery 8
7. Port Scanning 9
8. Service, Version and OS Detection 10
9. NSE Script Scanning 11
10. Firewall / IDS Detection 12
11. Export Nmap Scan Output 13
12. Scan Report 14
13. Analysis and Recommended Mitigations 15
14. Conclusion 16
15. Learning Experience 17
16. References 18

Evidence note

This version uses the screenshots supplied with the practical. The evidence is not a mockup. The host-discovery screenshot shows 10.204.133.156, while the later detailed scan screenshots show 10.118.110.18; this difference is retained rather than silently changed.

1. Introduction

Nmap (Network Mapper) is a network discovery and security auditing tool used to identify live hosts, discover network services, inspect ports, and collect information about exposed systems. In a practical assessment, Nmap can be used in stages: first establish reachability, then examine ports, identify services and versions, perform operating-system fingerprinting, run Nmap Scripting Engine (NSE) checks, and compare the results with filtering or firewall behaviour.

The supplied evidence shows Nmap 7.99 running on Kali Linux. The detailed scans against 10.118.110.18 reported the host as up but showed the scanned TCP ports as filtered/no-response. As a result, the supplied evidence does not establish any open service on that target.

The assessment should be performed only against an authorized laboratory system or network. Scanning an external system without permission can be intrusive and may violate policy or law.

Evidence used in this report

Seven terminal screenshots were supplied: Nmap command verification, host discovery, port scanning, service/version detection, OS detection, NSE script scanning, and ACK-based firewall/filtering detection.

2. Project Objective

The objective of this practical is to perform structured network reconnaissance using Nmap from Kali Linux. The practical covers the same stages described in the reference documentation:

• Verify the Nmap installation and command-line environment.

• Identify whether a target host is reachable.

• Scan TCP ports and determine their reported state.

• Attempt service and version identification.

• Attempt operating-system fingerprinting.

• Run default NSE scripts against the target.

• Use an ACK scan to observe firewall/filtering behaviour.

• Consolidate observations into a structured scan report.

The purpose is not simply to run commands, but to interpret the output and document what the scan actually established.

3. Scope

Target evidence

Host discovery evidence: 10.204.133.156. Detailed scan evidence: 10.118.110.18.

Assessment environment

Kali Linux terminal with Nmap 7.99. The screenshots show command-line execution using both the normal Kali user and a root terminal. No claim is made about the underlying target operating system because the supplied OS fingerprinting result was inconclusive.

Assessment boundaries

The practical is limited to discovery, scanning and observation. The supplied evidence does not demonstrate exploitation or unauthorized access.

Evidence limitation

The supplied screenshots do not include an exported XML/HTML Nmap report. Therefore, the export section provides the appropriate commands and explains the expected deliverables without claiming that an export was captured.

4. Tools Required

Component	Purpose
Kali Linux	Scanning workstation / security-testing environment
Nmap 7.99	Host discovery, port scanning, service/version, OS, NSE and filtering checks
Terminal	Execute Nmap commands and preserve command output
Authorized target	System or VM permitted for assessment
Optional: xsltproc	Convert Nmap XML output into a readable HTML report

Nmap verification evidence

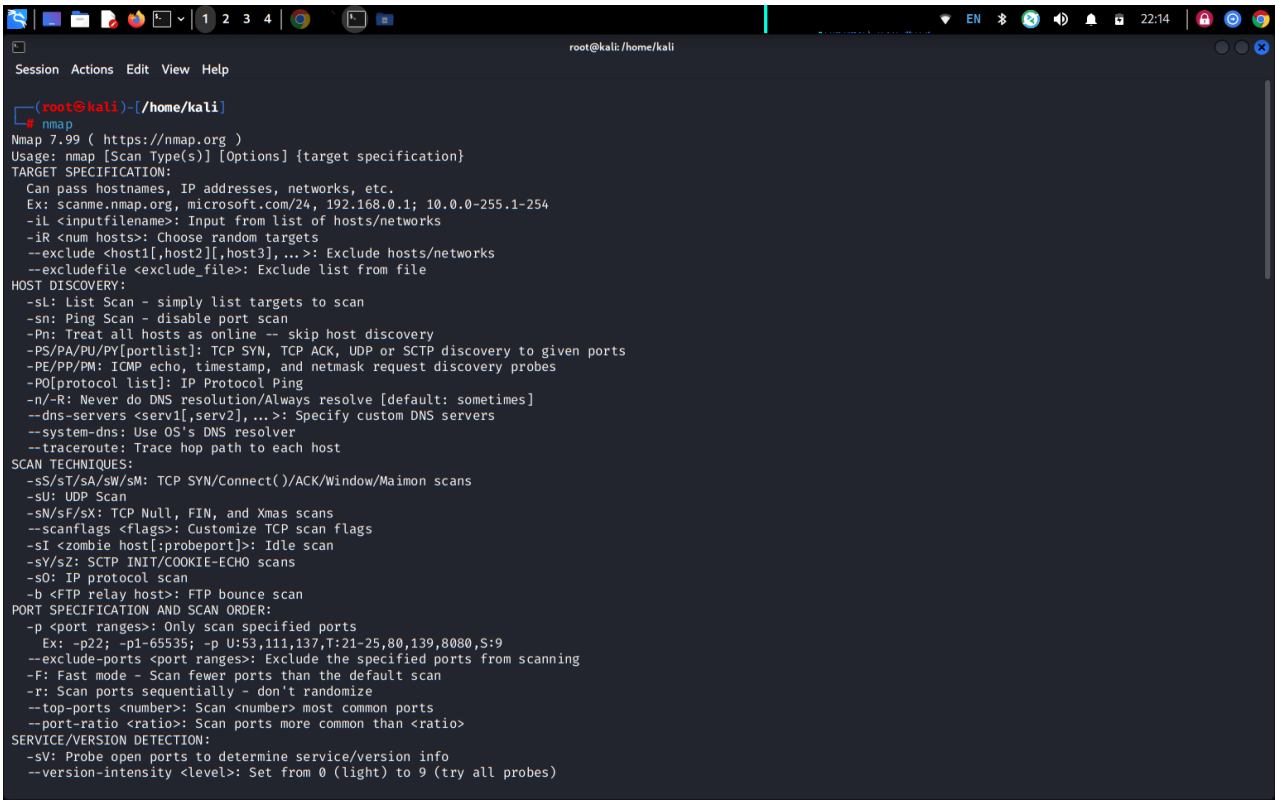


Fig. 1 — Nmap 7.99 command-line interface and available scan options.

5. Install Nmap and Verify Setup

Objective

To verify that Nmap is installed and available from the Kali Linux command line before performing network scanning.

Command used / observed

```
nmap
nmap --version
```

The supplied screenshot displays Nmap 7.99 and its usage information, confirming that the Nmap executable is available. The screenshot is a command-help output rather than a standalone version-only command.

Result

Nmap 7.99 was available and ready for the subsequent practical activities.

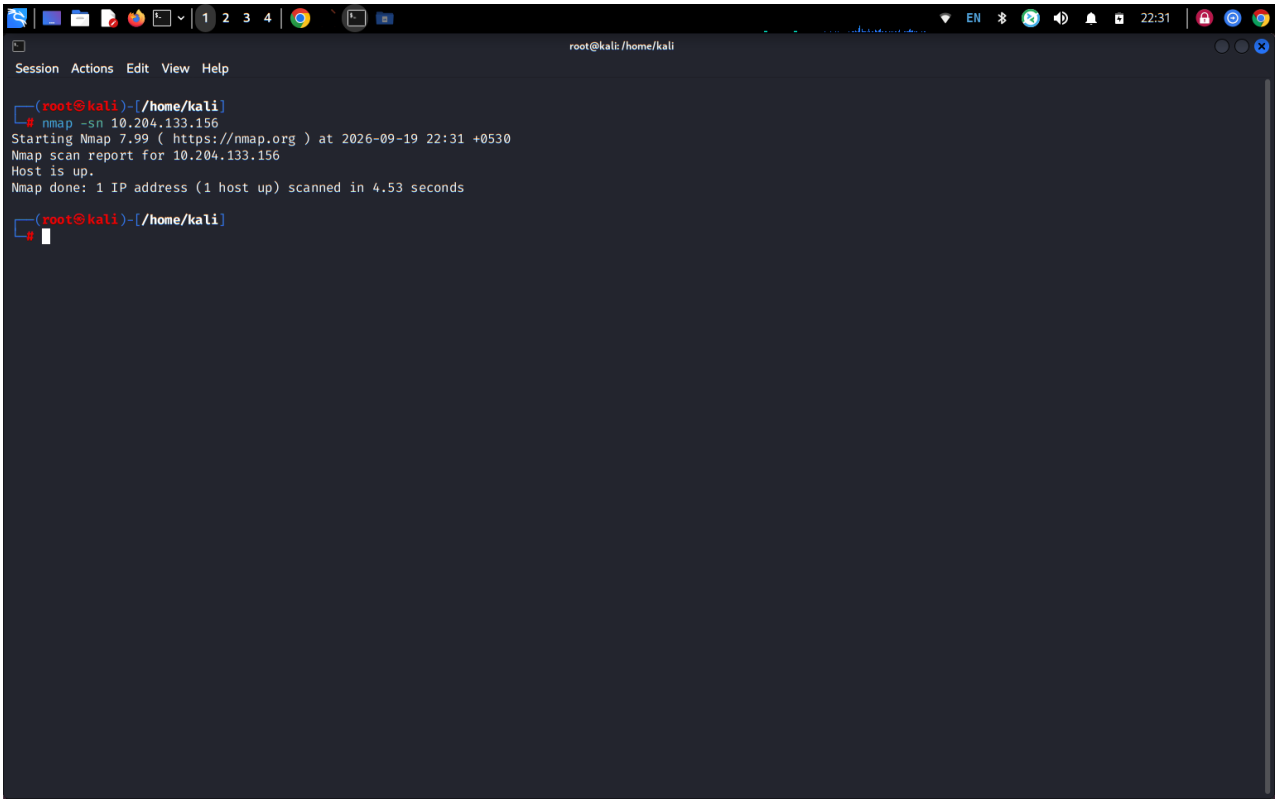


Fig. 2 — Host discovery using Nmap -sn; target 10.204.133.156 is reported as up.

6. Host Discovery

Objective

To determine whether a target host responds to network discovery probes before performing detailed port scanning.

Command

```
nmap -sn 10.204.133.156
```

The -sn option performs host discovery without a normal port scan. In the supplied evidence, Nmap reports the target as up and completes the scan of one IP address in 4.53 seconds.

Observed result

10.204.133.156 was reported as up in the supplied host-discovery capture.

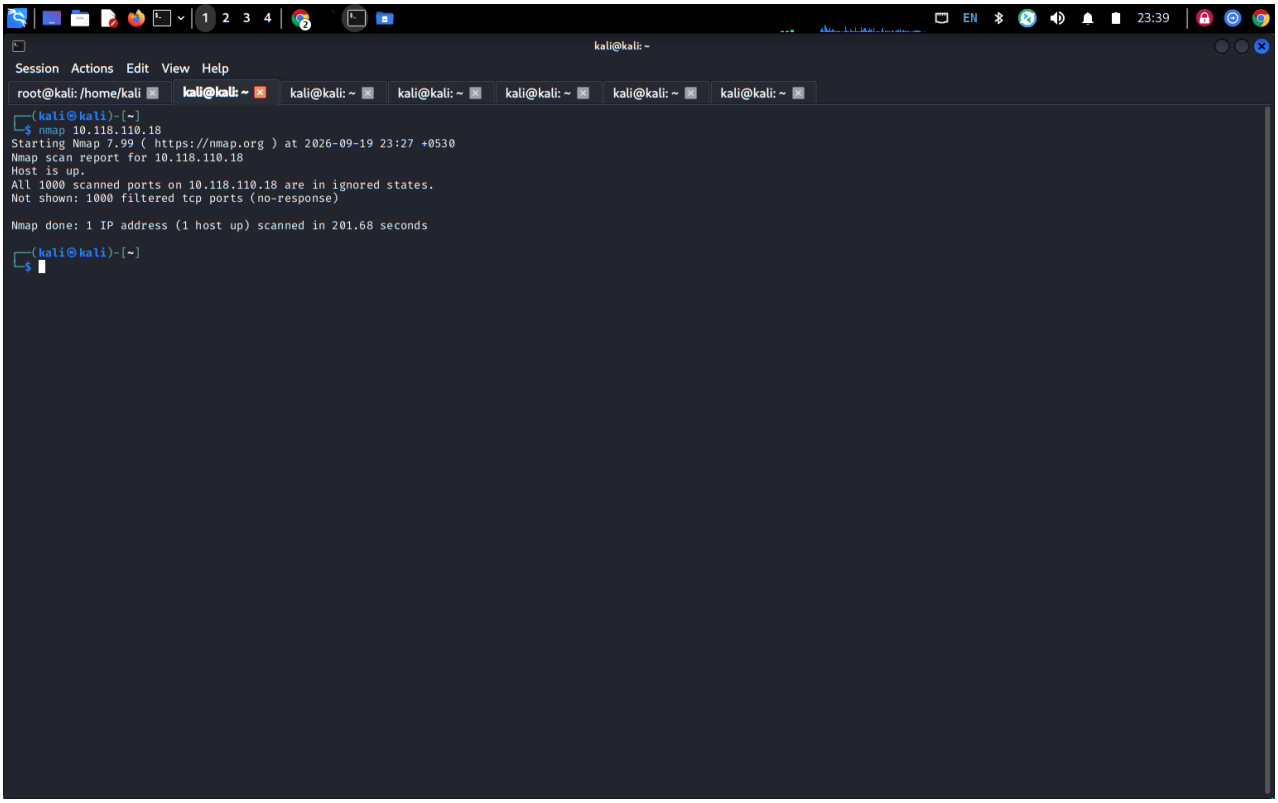


Fig. 3 — Default TCP port scan of 10.118.110.18; 1000 ports are reported as filtered/no-response.

7. Port Scanning

Objective

To determine the state of TCP ports on the target and identify whether services are exposed.

Command

```
nmap 10.118.110.18
```

The supplied capture shows the default Nmap scan completing against 10.118.110.18. Nmap reports that all 1000 scanned TCP ports are in ignored states and that 1000 ports are filtered with no response.

Observed result

No open TCP ports are shown in the supplied screenshot. The result is consistent with filtering/no-response from the target for the default TCP port set.

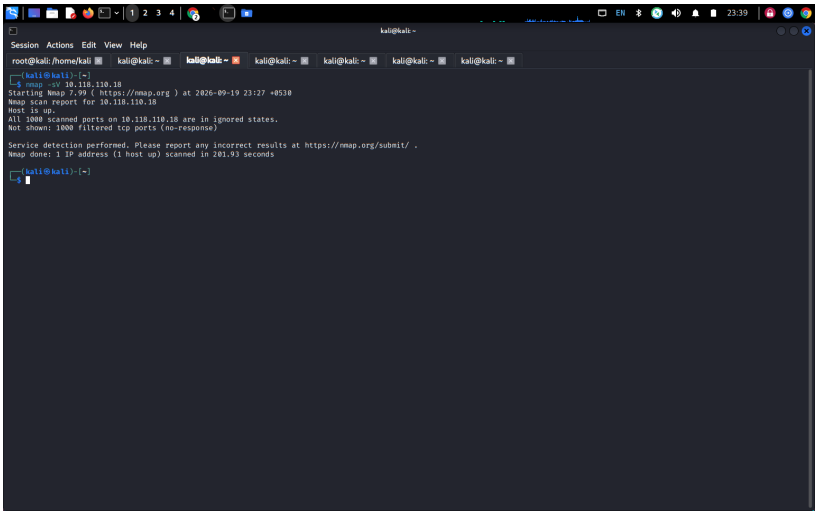


Fig. 4 — Service/version detection attempt against 10.118.110.18; no open services are reported.

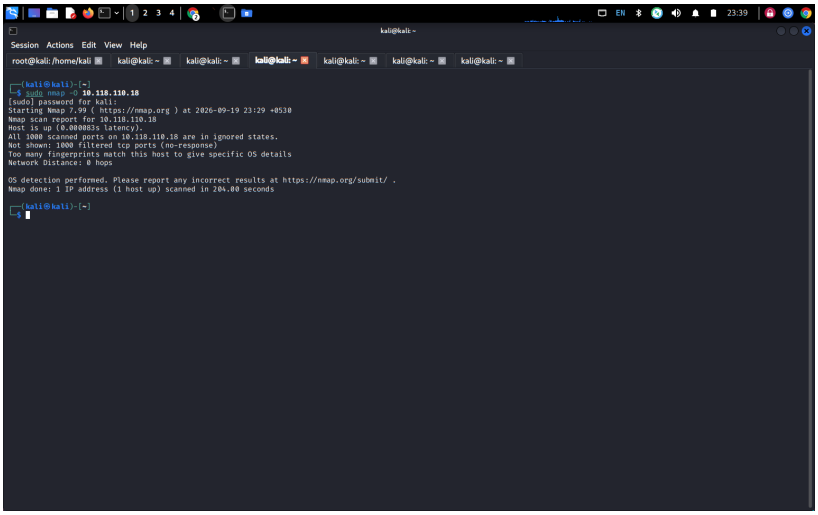


Fig. 5 — OS detection is inconclusive because multiple fingerprints match.

8. Service, Version and OS Detection

Service and version detection

```
nmap -sV 10.118.110.18
```

The -sV scan attempts to identify services and versions on discovered open ports. In the supplied capture, all 1000 scanned ports are reported as filtered/no-response, so no service/version banner is identified.

Operating-system detection

```
sudo nmap -O 10.118.110.18
```

The OS scan reports that too many fingerprints match to provide specific OS details. Network distance is reported as 0 hops.

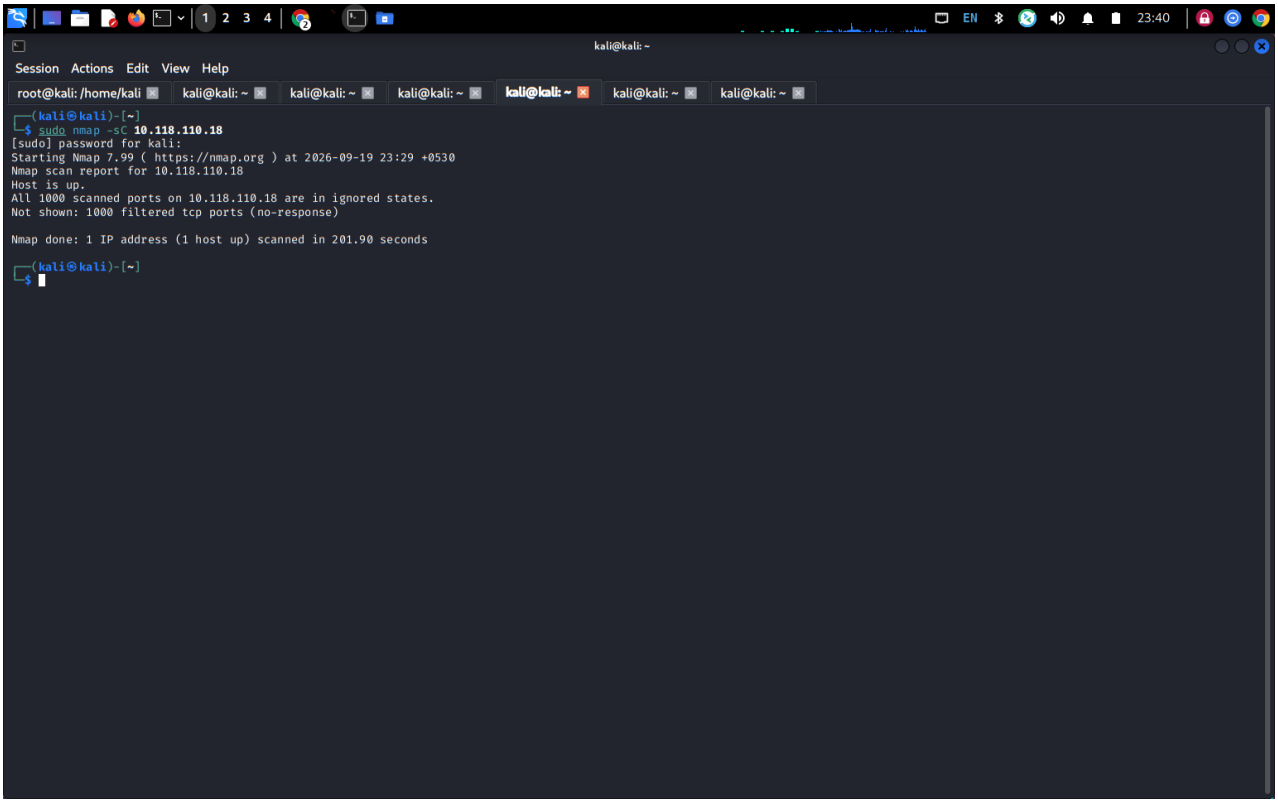


Fig. 6 — Default NSE script scan against 10.118.110.18; no script findings are shown in the supplied output.

9. NSE Script Scanning

Objective

To run Nmap Scripting Engine checks and observe whether default scripts produce additional information about the target.

Command

```
sudo nmap -sC 10.118.110.18
```

The -sC option runs Nmap's default script set. In the supplied evidence, the target is up but all 1000 scanned TCP ports are filtered/no-response. No NSE findings are displayed in the captured output.

Result

The supplied screenshot does not provide evidence of an NSE-detected vulnerability or misconfiguration. This should not be interpreted as proof that the host is vulnerability-free; it only describes what this scan established.

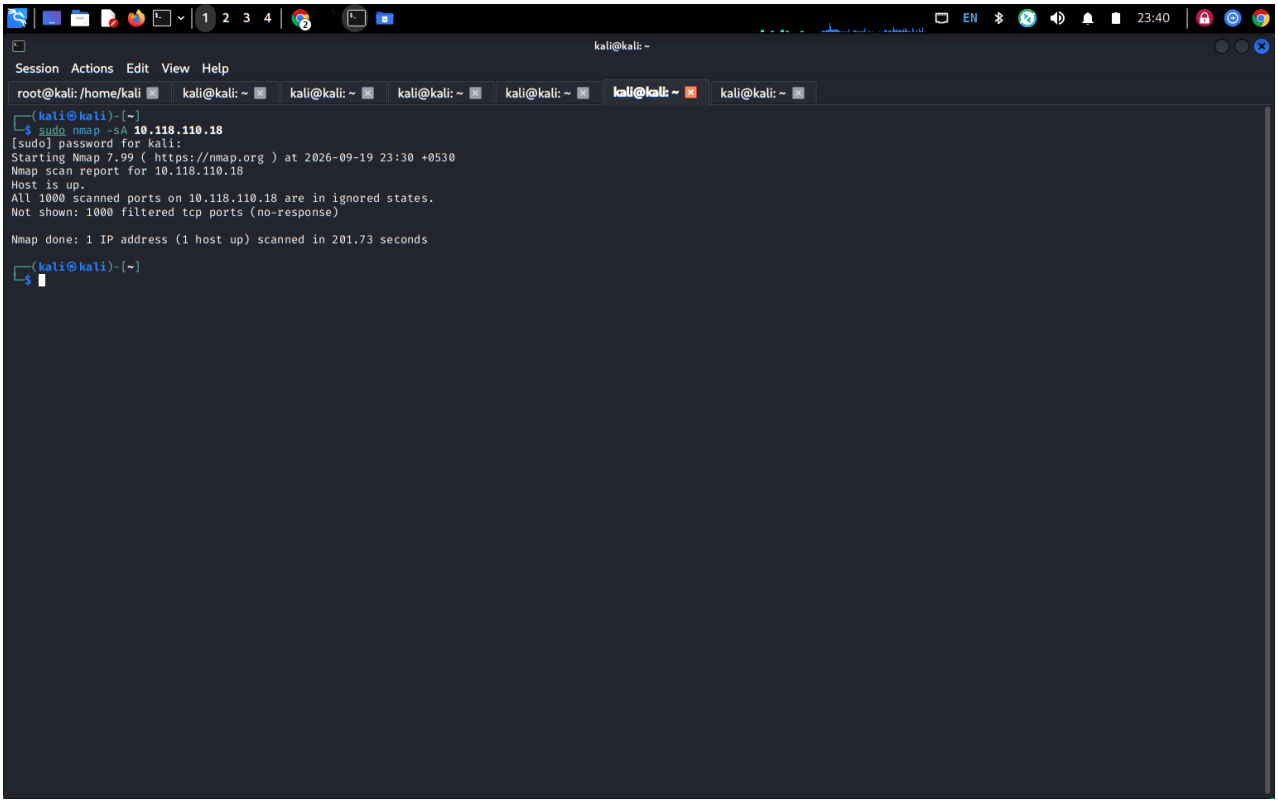


Fig. 7 — ACK scan against 10.118.110.18; all 1000 scanned ports are reported as filtered/no-response.

10. Firewall / IDS Detection

Objective

To observe whether TCP filtering behaviour is present by using an ACK scan and comparing the result with the normal TCP scan.

Command

```
sudo nmap -sA 10.118.110.18
```

An ACK scan is commonly used to help characterize filtering. In the supplied capture, all 1000 scanned ports are reported as filtered/no-response.

Observed result

The evidence indicates filtering/no-response behaviour. It does not identify the exact firewall product or prove that a specific IDS is present.

11. Export Nmap Scan Output

Objective

To save the final assessment in multiple formats so the results can be archived and reviewed.

Recommended command for an authorized target

```
nmap -A -T4 -oA final_scan 10.118.110.18
```

The -oA option writes three related output files:

• final_scan.nmap — normal text output
• final_scan.xml — structured XML output
• final_scan.gnmap — grepable output

HTML conversion (if xsltproc is installed)

```
xsltproc final_scan.xml -o final_scan.html
firefox final_scan.html
```

Evidence status

No exported XML/HTML screenshot was supplied with the evidence package. Therefore this section documents the reproducible export procedure rather than claiming an export was completed.

12. Scan Report

The following table consolidates only the results visible in the supplied screenshots.

Stage	Target	Observed result	Interpretation
Host discovery	10.204.133.156	Host is up	Reachability confirmed for this capture
Port scan	10.118.110.18	1000 TCP ports filtered/no-response	No open ports shown
Service/version	10.118.110.18	Service detection performed; no open ports shown	No service/version identified
OS detection	10.118.110.18	Too many fingerprints; no specific OS	OS identification inconclusive
NSE default	10.118.110.18	No script findings shown	No additional findings in supplied output
ACK/firewall check	10.118.110.18	1000 ports filtered/no-response	Filtering behaviour observed

Key observation

For 10.118.110.18, the supplied scans consistently show the host as reachable while the default TCP port set is filtered/no-response. Consequently, the evidence does not support listing an open service, version, or confirmed operating system for this target.

13. Analysis and Recommended Mitigations

Analysis of the supplied evidence

The most consistent observation is filtering of the default TCP port set on 10.118.110.18. From a defensive perspective, filtering reduces direct exposure, but the screenshots alone cannot determine which firewall rule, gateway, host firewall, or network security device produced the behaviour.

Recommended defensive measures

• Expose only services that are required for the intended function of the system.

• Use host and network firewalls to restrict unnecessary inbound traffic.

• Review filtered/allowed ports periodically against an approved asset and service inventory.

• Keep operating systems and exposed services patched and supported.

• Use logging and monitoring to identify repeated scanning or unusual connection attempts.

• Use vulnerability assessment tools such as Nessus in addition to Nmap when CVE/CVSS-oriented assessment is required.

• Repeat authorized scans after firewall or service changes to verify the intended attack surface.

These are general defensive recommendations; they are not claims that a specific vulnerability was found in the supplied scan evidence.

14. Conclusion

The Nmap practical was completed through a sequence of host discovery, TCP port scanning, service/version detection, OS fingerprinting, NSE script scanning, and ACK-based filtering analysis.

The supplied host-discovery evidence showed 10.204.133.156 as reachable. The subsequent detailed evidence for 10.118.110.18 showed the host as up, while the default 1000 TCP ports were reported as filtered/no-response. Service/version detection did not identify an open service, OS detection was inconclusive, and the default NSE scan displayed no findings in the captured output.

The practical demonstrates an important security-assessment principle: a scan result must be interpreted according to what the evidence actually shows. A filtered port is not the same as an open port, and an inconclusive OS fingerprint is not an identified operating system.

15. My Learning Experience

This practical helped me understand how Nmap can be used systematically rather than as a single command. The sequence of discovery, port scanning, service detection, OS detection, NSE scripts, and filtering analysis makes it easier to understand what information each stage contributes.

One important observation from my screenshots was that the target 10.118.110.18 was reachable but returned filtered/no-response results for the scanned TCP ports. I also learned that service/version and OS detection depend on the information available from the target; when ports are filtered, identification can become limited or inconclusive.

I also learned the importance of keeping evidence consistent and recording the exact target address used for each command. The supplied screenshots contain two different target addresses, so the final documentation records that difference instead of treating the captures as if they came from one host.

16. References

The reference documentation supplied for this practical was used for the organization, terminology, and sequence of sections.

• Nmap Official Documentation — https://nmap.org/book/man.html

• Nmap Scripting Engine (NSE) — https://nmap.org/book/nse.html

• Metasploitable2 Documentation — https://docs.rapid7.com/metasploit/metasploitable-2/

• Common Vulnerabilities and Exposures (CVE) Database — https://cve.mitre.org

End of Practical Documentation

Network Scanning Using Nmap — Kali Linux